QUIZ 08, PRACTICE PROBLEMS – Solutions

STAT 1110: Section – Fall 2026 with Dr. Ethan P. Marzban

⚠ Caution

These problems are not meant to be exactly indicative of what will show up on the Quiz on Monday, but are rather designed to be a starting point for your studies.

Some General Notes for this Quiz:

- The quiz will cover paired data (Chapter 7), and a simulation-based approach to testing multiple proportions (Section 8.1)
- There will be a handful of multiple-choice and open-ended questions

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Scenario 1: For Multiple Choice Questions 1 - 5 and Free-Response Questions 1 - 2.

A (fictional) study recruited 150 adult volunteers aged 18 - 28 to undergo a study that lasted for one night, designed to test the effectiveness of three different **pre-bedtime routines**: scrolling through your phone, listening to soothing music, and doing nothing (and going straight to bed). These volunteers were randomly split into three groups: one group was instructed to scroll through their phones right before bed, another group was instructed to listen to soothing music before bed, and the third group was instructed to do nothing before bed (and just try to go straight to sleep). The next morning, the volunteers self-reported **whether they felt they got a good night's sleep or not** ("good nights sleep" is a "success"). Our research question is: **is there an association between the pre-bedtime activity and whether or not you get a good night's sleep?**

Here is a summary of the data:

	Phones	Music	Nothing	Total
Good Sleep	25	40	30	95
Total	50	50	50	150

Table 1: Summary of Ayoade's data

MULTIPLE CHOICE QUESTION 1

What is one observational unit in this study?

(**A**) One person who got a good night's sleep

(**B**) One person who did not get a good night's sleep

(C) One person

(**D**) One pre-bedtime routine

FREE-RESPONSE QUESTION 1

Write down the null and alternative hypotheses in words and, if possible, symbols and math. Be sure to define any symbols you use!

Solution: In Words:

- H_0: there is no association between bedtime routine and whether or not you get a good night's sleep
- H_A: there is an association between bedtime routine and whether or not you get a good night's sleep

Let π_{Phones} denote the long-run proportion of times someone in the "phone" bedtime routine got a good night's sleep; let π_{Music} denote the long-run proportion of times someone in the "music" bedtime routine got a good night's sleep; let π_{None} denote the long-run proportion of times someone in the "none" bedtime routine got a good night's sleep. Then, in symbols:

- H_0: $\pi_{\text{Phones}} = \pi_{\text{Music}} = \pi_{\text{None}}$
- There is no way to write the alternative in symbols and math only.

Some points I would grade on:

- Making sure you had the right verbal hypotheses
- Making sure you defined your parameters correctly (identifying "proportion;" including either "long-run" or "population;" identifying that we need three long-run proportions, one for each group)
- Making sure you identified there is no way to write the alternative in symbols and math only

FREE-RESPONSE QUESTION 2

Calculate the value of the mean group difference statistic.

Solution: We start by calculating the pairwise differences in sample proportions:

Groups Diff.	Difference in Sample Proportions
Phones - Music	$\frac{25}{50} - \frac{40}{50} = -0.3$
Phones – Nothing	$\frac{25}{50} - \frac{30}{50} = -0.1$
Music - Nothing	$\frac{40}{50} - \frac{30}{50} = 0.2$

Table 2: Partial Calculations

The Mean Group Diff statistic value is the average of the absolute values of these pairwise differences:

$$\text{Mean Group Diff} = \frac{1}{3}\left(|-0.3| + |-0.1| + |0.2|\right) = \frac{1}{3}(0.6) = \mathbf{0.2}$$

MULTIPLE CHOICE QUESTION 2

True or False: larger values of the Mean Group Diff statistic correspond to weaker evidence against the null hypothesis.

(**A**) True **(B) False**

Solution: Remember: values of the Mean Group Diff statistic that are closer to zero provide weaker evidence against the null; mean group diff statistic values that are *large* correspond to *stronger* evidence against the null.

FREE-RESPONSE QUESTION 3

Describe the shape of the null distribution.

Solution: The null distribution, when using the Mean Group Diff statistic, will be unimodal and right-skewed. **Remember:** when describing the shape of a distribution, you must make reference to the number of peaks and whether or not it is symmetric (and, if it is asymmetric, which direction the skew is)

MULTIPLE CHOICE QUESTION 3

After running 10,000 repetitions in the Multiple Proportions applet, the experimenters obtained a simulation-based p-value of 0.0059. What is the strength of evidence this data provides against the null hypothesis?

(A) Very strong evidence against the null hypothesis

(**B**) Strong evidence against the null hypothesis

(**C**) Moderate evidence against the null hypothesis

(**D**) Not much evidence against the null hypothesis

In addition to the original setup, use the following for Free Response Question 4 and Multiple Choice Question 4 Jonah, a STAT 1110 student, wants to conduct one additional repetition of the simulation. He knows he needs to use blue cards (for "successes") and red cards (for "failures"), but can't remember much beyond that.

FREE-RESPONSE QUESTION 4

How many index cards of each color does Jonah need?

Solution: 95 blue (for the 95 people who got a good night's sleep) and 55 red (for the 150 – 95 = 55 people who did not get a good night's sleep).

MULTIPLE CHOICE QUESTION 4

After selecting the appropriate number of each color card, Jonah will shuffle them up well. Which of the following best describes what the shuffling seeks to accomplish?

(**A**) The shuffling models taking a simple random sample, which will enable us to generalize our results to a broader population.

(**B**) The shuffling models random assignment, which will enable us to make causal claims

(C) The shuffling breaks any potential association between the explanatory variable and response variable, allowing us to conduct our simulation "under the null"

Multiple Choice Question 5 is unrelated to the previous problems.

MULTIPLE CHOICE QUESTION 5

Recall from lecture: if there is a 5% chance I fall down while skiing down a slope, then the chances I fall at least once among 10 times I ski down the slope are around 40.13%. Which of the following is the best/most correct interpretation of what this means in the context of the **multiple testing problem?**

(**A**) Using a 5% significance level, the probability of committing a Type I error among 10 hypothesis tests is 40.13%

(B) Using a 5% significance level, the probability of committing at least one Type I error among 10 hypothesis tests is 40.13%

(**C**) Using a 5% significance level, the probability that at least one null hypothesis out of 10 tests being conducted on the same dataset is 40.13%

(**D**) Using a 5% significance level, the probability that the mean group diff statistic is greater than or equal to 1 is 40.13%

Scenario 2: For Multiple Choice Question 6 and Free-Response Questions 5 - 6.

In a previous lecture activity ("Plate Up!") we explored the potential association between the size of the plate on which food is served and the amount of calories (in kCal) that are consumed.

Jasmine has conducted a follow-up study. She recruited 10 volunteers, and randomly divided them into two groups of 5. One group was served food on a small plate, the other was served food on a big plate. However, unlike the original study (in which all participants were served the same food) Jasmine was unable to provide the same food to all volunteers. As such, she decided to match volunteers between the two groups who had similar food served.

As a reminder, the research question is: **does the size of plate on which food is served affect the amount of calories of the food that are consumed?**

Whenever necessary, use the order (Big – Small).

MULTIPLE CHOICE QUESTION 6

Why has Jasmine decided to match people between the two groups?

(**A**) Because this was an observational study, there is the potential for confounding effects. Matching based on food type helps control against potential confounding effects due to food type.

(**B**) Because Jasmine relied on volunteers, she has a limited scope of inference. Matching based on food type will allow her to generalize her results more broadly than if she hadn't matched.

(C) Even though this is a randomized experiment, and randomization balances with respect to potential confounding variables on average, any one particular assignment may exhibit noticeable imbalances. Matching based on food type, which is probably associated with the amount of calories consumed, ensures Jasmine can more readily attribute any differences in calories consumed between the two groups to the plate size and *not* food type.

FREE-RESPONSE QUESTION 5

Write down the null and alternative hypothesis in symbols and math only, if possible. If it is not possible to do so, explain why. Be sure to define any symbols you use!

Solution: Because this is a paired design, we define just one symbol: the population/long-run average difference. More specifically, we define μ_d to be the long-run average difference (using the order of Big – Small) in calories consumed (in kCal), so that the null and alternative hypotheses become:

$$\begin{cases} H_0: & \mu_d = 0 \\ H_A: & \mu_d \neq 0 \end{cases}$$

FREE-RESPONSE QUESTION 6

Jasmine has calculated an approximate 90% confidence interval for the parameter of interest to be **(12.03, 15.04)**. Interpret this interval in context.

Solution: We are 90% confident that the long-run average amount of calories consumed by people who have food served to them on a big plate is between 12.03 kCal and 15.04 kCal HIGHER than the long-run average amount of calories consumed by people who have food served to them on a small plate, for volunteers similar to those in Jasmine's study.

Some things I would look for when grading a problem like this:

- Using wording like "90% confident"
- Indicating which is HIGHER or LOWER
- Including "for volunteers similar to those in Jasmine's study"